Quantifying Uncertainty in Tree Biomass — Student Guide

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This guide shows how to put an honest error bar on the biomass, carbon and CO_2 of a tree you measured once, and on the total of your 25 trees. The short answer: one tree is uncertain by about a factor of two; the plot total by about ±12 %; and most of that comes from the equation, not from your tape.

How to use this guide

- **For the project:** read Parts 4–7 and follow the checklist in 8.2. Each step of the checklist points back to the part that explains it.
- **For the exam:** Parts 1–3 are the concepts; 8.3 has practice questions with answers; 8.4 is the glossary.
- **If statistics feels rusty:** start at Part 1 and do not skip 1.4 (the $\sqrt{n}$ rule) and 1.9 (how errors combine). Everything else is these two ideas applied.
- **Interactive pages:** four short web pages let you watch the methods work. Open them on a phone or laptop; no account is needed. Each part names its page.

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Interactive pages

Page	What you will see	Goes with
Dice to Bootstrap	a die rolled 10,000 times; averages of k dice narrowing by √k; the 24 height readings of tree #406 drawn from a bag with replacement, 10,000 times, until a 95 % interval appears	Parts 1–2
How an Equation Is Born	trees felled, weighed and plotted; the best line drawn; refitting on new samples (the line barely moves); the scatter band that never shrinks	Part 3
Monte Carlo, One Tree	five dice (H, D, ρ, parameters, scatter) rolled one draw at a time; the pile of 20,000 answers; one die at a time to find the culprit	Parts 4–6
From 25 Trees to a Plot	each tree's ±40 % whisker; the total built draw by draw to ±12 %; the error-versus-n curve; per m² and per hectare with the warnings	Part 7

Each page works on a phone, has Play / Step / Fast-forward buttons, and lets you change the assumed uncertainties to see what happens. All numbers on them are illustrative; replace them with your own data for the project.

Part 1 · Revision: the little statistics you need

You need only six ideas: mean, standard deviation, standard error, the normal curve, the 95 % interval, and how errors combine. Each is explained with one running example, the 24 height readings of tree #406.

1.1 Population, sample, and why we never see the truth

The **true** height of tree #406 is one number. Nobody knows it. What we have is a **sample**: 24 groups each read a height, and the readings differ (5.4 m, 6.3 m, 5.9 m, …). Statistics is the craft of saying something honest about the truth from a sample that disagrees with itself.

1.2 Mean: the centre of the pile

Add the values, divide by how many there are.

```
\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n}
```

For the 24 heights the mean is about 6.15 m. The mean is our best single guess of the truth.

1.3 Standard deviation (SD): the typical distance from the centre

Take each reading, subtract the mean, square it (so negatives do not cancel positives), average the squares, take the square root.

```
s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}
```

For the 24 heights, s ≈ 0.65 m. Read it as: *one reading, picked at random, is usually about 0.65 m away from the mean*. A small SD means readers agree; a large SD means they do not. The **coefficient of variation** CV = s / x̄ writes the same spread as a percentage (0.65 / 6.15 ≈ 11 %), which lets you compare the wobble in height (metres) with the wobble in diameter (centimetres).

1.4 Standard error (SE): how sure we are about the mean

The SD tells you how much *one reading* wobbles. But we do not report one reading, we report the mean of 24. The mean wobbles much less, because high and low readings partly cancel. The rule:

```
SE = \frac{s}{\sqrt{n}}
```

For tree #406: SE = 0.65 / √24 ≈ 0.13 m. Doubling the number of readings does not halve the error; you need **four times** as many readings to halve it. This √n rule is the single most important formula in this guide. It returns in Part 7 when we add 25 trees.

1.5 The normal (bell) curve and the 68–95 rule

When many small, independent causes push a measurement up or down, the pile of readings takes a bell shape. For a bell curve:

- about 68 % of values fall within ±1 SD of the mean,
- about 95 % fall within ±2 SD (more exactly ±1.96 SD),
- about 99.7 % fall within ±3 SD.

You do not need the formula of the curve. You need the 95 % rule, because that is what a 95 % confidence interval is built from.

1.6 The 95 % confidence interval (CI)

Two ways to get it. They agree when the pile is bell-shaped.

Formula way: mean ± 1.96 × SE. For tree #406: 6.15 ± 1.96 × 0.13 → **5.90 to 6.40 m**.

Counting way (no formula): make a long list of possible values of the mean (Part 2 shows how), sort it, and cut off the lowest 2.5 % and the highest 2.5 %. What remains, the middle 95 %, is the interval. This way also works when the pile is lopsided, which biomass piles always are.

How to read a 95 % CI: *if we repeated the whole exercise many times, about 95 of every 100 intervals built this way would contain the true value*. It is a statement about the method, not a promise about this one tree.

1.7 Lopsided piles: the log-normal

Biomass errors are **multiplicative**: a tree is 1.4 times heavier than predicted, or 1.4 times lighter, not "+5 kg" or "−5 kg". A pile of multiplicative errors is squashed on the left and stretched on the right (you cannot be 150 % too light, but you can be 150 % too heavy). Taking the logarithm turns "times" into "plus" and makes the pile bell-shaped again. That is why the Chave equation is fitted on log axes and why its scatter σ = 0.357 is quoted in log units: e^0.357 ≈ 1.43, so "±40 %" roughly.

1.8 Random versus systematic error

- **Random error** scatters both ways around the truth with no pattern. Averaging many readings shrinks it (by √n).
- **Systematic error (bias)** leans the same way every time, like a bathroom scale that always reads 2 kg heavy. Averaging does **not** shrink it. Only a better instrument, a better model or a correction removes it.

In this guide the model residual scatter and person-to-person wobble are random; the wood-density table value, the carbon fraction and the equation's +5 % lean are systematic.

1.9 How errors combine (propagation)

When a result is built from several uncertain inputs, the uncertainties do not simply add. For **independent** random errors you square each, add, and take the square root ("add in quadrature"):

```
\sigma_{total} = \sqrt{\sigma_1^2 + \sigma_2^2 + \sigma_3^2 + \dots}
```

Example: 37 % (model) and 10 % (height) and 5 % (diameter) combine to √(37² + 10² + 5²) ≈ 38.6 %, not 52 %. The biggest term dominates; the small ones almost vanish.

For a **product or quotient** the *relative* (percent) errors add in quadrature. For a **power**, multiply the relative error by the exponent: D² carries twice the percent error of D, so a 3 % wobble in D becomes a 6 % wobble in D². For a sum of **independent** quantities (25

trees), the *absolute* errors add in quadrature, which is exactly why the total's percent error shrinks like 1/√n.

These rules are all you need to check a computer result by hand. The computer methods in Part 2 do the same job without any formula and without assuming independence or bell shapes.

Part 2 · Two computer tricks that replace formulas: bootstrap and Monte Carlo

Both tricks rest on one idea: **probability is a long-run frequency**. If you can repeat something many times, or make a computer repeat it, you can *count* how often each outcome happens instead of deriving it. Interactive page: *Dice to Bootstrap* (link in the table at the top of this guide).

2.1 The analogy: a die, rolled many times

Roll a fair die 10,000 times. Each face comes up about 1,667 times, never exactly. Watch the running average: after 10 rolls it may be 2.9 or 4.1; after 10,000 rolls it sits within a few hundredths of 3.5. The wobble of the average shrinks like 1/√n, the rule from Part 1.4.

Now average two dice per throw: the pile of averages is narrower (SD 1.21 instead of 1.71). Average ten dice: narrower still (SD 0.54 = 1.71/√10). Nothing about a die changed; averaging did the work.

A measurement is a roll of a die you did not design. The die is the whole process: the person, the tape, the angle reader, the canopy, the light. One group's reading of tree #406 (say 6.1 m) is one roll. The faces are all the heights that process could produce, loaded in a way nobody chose. We have 24 rolls, one per group. Can we learn the shape of the die from only 24 rolls?

2.2 Bootstrap: when you have data but no formula

Yes, by re-rolling the 24 we already have. This is the **bootstrap** (Efron, 1979; the "bag of chips" picture is from Efron & Tibshirani, 1993).

1. **Bag.** Write each of the 24 height readings on a chip and put the chips in a bag.
2. **Draw.** Pull one chip, write down its value, **put it back**. Repeat 24 times. You now have a pretend sample of 24 readings. Some originals appear twice or three times; some do not appear at all.
3. **Compute.** Take the mean of the pretend sample.
4. **Repeat.** Do steps 2–3 ten thousand times. You get 10,000 pretend means.

5. **Read off.** The SD of the 10,000 means is the standard error. Sort them and cut 2.5 % off each end: the middle 95 % is the confidence interval.

For tree #406 (24 readings, mean 6.14 m, SD 0.68 m) the bootstrap gives SE ≈ 0.13 m and a 95 % interval of about **5.88 to 6.40 m**. The formula s/√n gives 0.138 m. They agree. No formula was needed, only counting.

Why put the chip back? We are pretending the 24 chips *are* the die. A die can show the same face twice, so a chip must be allowed to come out twice. Drawing without replacement would just reshuffle the same 24 values and every pretend mean would be identical.

Assumptions the bootstrap makes (say them in your report):

- The 24 readings are a fair sample of what the measurement process can produce. If every group used the same wrong eye height, the bootstrap cannot see that bias.
- The readings are independent of each other. Groups did not copy each other's numbers.
- 24 is enough to sketch the shape of the die. With 5 readings the sketch is crude; with 3 it is useless.
- The bootstrap tells you about **precision** (wobble), never about **accuracy** (lean). A perfectly repeatable wrong method bootstraps to a tiny interval around a wrong answer.

2.3 Monte Carlo: when you know the dice and want the result

The bootstrap learns a die from data. **Monte Carlo** goes the other way: you already know (or assume) the shape of each die and want to know how their combination behaves. The name comes from the casino; the method was born in 1940s physics when the formulas became too hard.

For biomass the "dice" are the uncertain inputs: height H, diameter D, wood density ρ, the two fitted numbers of the equation, and the model's leftover scatter ε. The recipe:

1. **Roll every die once.** Pick one plausible H, one D, one ρ, one pair (a, b), one ε, each from its own distribution.
2. **Compute.** AGB = a · (ρ · D² · H)^b × e^ε.
3. **Write the answer down.**
4. **Repeat** 20,000 times.
5. **Read off.** The middle of the pile (median) is the estimate; the SD or CV is the uncertainty; the middle 95 % of the pile is the interval.

Where do the dice come from? From data when you have it (the bootstrap of 24 readings gives the H die), from the literature when you do not (σ = 0.357 for the model scatter, four published values for ρ), and from an **educated guess** when neither exists (±20 % for height measured once). Part 6 lists each die for the Chave equation.

2.4 Which trick when?

Situation	Use	Example in this project
You have repeated readings and want the uncertainty of their mean	Bootstrap	24 group readings of tree #406; 10 reference trees measured by every team
You have one reading and a guess at its wobble	Monte Carlo with an assumed distribution	H measured once, SD assumed 10 %
You know several uncertain inputs and want the uncertainty of a *computed* result	Monte Carlo	AGB of one tree; total of 25 trees
You want to know **which** input matters most	Monte Carlo, one die at a time	Part 5.3

2.5 How many repeats?

The wobble of a Monte Carlo result itself shrinks like 1/√N, where N is the number of repeats. With N = 10,000 the 95 % interval limits are known to about ±1 %. A laptop does 20,000 repeats of the biomass equation in well under a second, so N is never the bottleneck. Use at least 5,000; use 20,000 for the report.

Part 3 · Where a biomass equation comes from, and the two ways it can be wrong

You cannot weigh a living tree, so you predict its weight from its shape. Every such prediction carries two separate model uncertainties: the fitted numbers could be slightly off (small), and real trees scatter around any line (large). Interactive page: *How an Equation Is Born*.

3.1 Cut, dry, weigh

An **allometric equation** predicts something hard to measure (dry weight) from things easy to measure (diameter D, height H, wood density ρ). To build one, researchers:

1. Choose trees of many sizes and species.
2. Measure D and H standing; take a wood sample for ρ.
3. Fell each tree, cut it up, weigh everything above ground (trunk, branches, leaves).
4. Oven-dry sub-samples to convert fresh weight to **dry weight** (water is not biomass).
5. Plot dry weight against size and fit a curve.

Chave et al. (2014) assembled 4,004 such trees from 58 sites across the tropics, from 5 cm saplings to 100-tonne giants, and fitted one equation to all of them:

```
AGB = 0.0673 \times (\rho \, D^2 \, H)^{0.976}
```

AGB in kg of oven-dry matter, ρ in g/cm³, D in cm, H in m. The bracket ρ·D²·H is roughly the mass of a cylinder as wide and tall as the tree; a real tree weighs about 7 % of that cylinder because it tapers and branches. The two numbers 0.0673 and 0.976 are not theory. They are the intercept and slope of a straight line fitted through 4,004 points on log–log axes (logs turn "times" into "plus" and make the scatter bell-shaped, Part 1.7).

3.2 Uncertainty 1 — parameter uncertainty: where exactly is the line?

Harvest a *different* 4,004 trees and refit, and you get slightly different numbers, say 0.0681 and 0.975. The fitted parameters are themselves estimates with a standard error. On the interactive page, refitting on a fresh sample of 50 trees moves the line visibly; refitting on 4,000 trees barely moves it. With 4,004 trees behind it, the Chave line is pinned down to well under 1 % for a tree the size of ours.

Two properties matter:

- It is **systematic**: if the true slope is 0.980 rather than 0.976, *every* tree you compute is off in the same direction. Measuring more trees does not cancel it.
- It is **small** here, so in this project we carry it as a shared ±1 % and move on.

3.3 Uncertainty 2 — residual scatter: trees do not sit on the line

Even with a perfect line, real trees land above or below it. Two trees with identical D, H and ρ do not weigh the same: one has more branches, one lost its top years ago. The vertical spread of the cloud around the line is the **residual standard error**, written σ (sigma). For the Chave equation σ = 0.357 in natural-log units.

What 0.357 means in plain numbers: e^0.357 = 1.43. So for one tree the true weight is usually (68 % of the time) between 0.7× and 1.4× the prediction, and 95 % of the time between 0.5× and 2.0× (e^(±1.96×0.357)). For tree #406, predicted 13.6 kg, that is roughly 7 to 28 kg. This is **by far the biggest single uncertainty in the whole project**.

Two properties, the opposite of Uncertainty 1:

- It is **random**: each tree lands independently above or below. Add 25 trees and the highs and lows partly cancel (Part 7).
- It **does not shrink** when more trees are harvested. It is a property of trees, not of the sample size. A million felled trees would pin the line down perfectly and the cloud would be exactly as wide.

3.4 Side by side

	Parameter uncertainty	Residual scatter
Question	Where is the best line?	How far is one tree from the line?
Size for our trees	< 1 %	about ±40 % (1 SD), ± factor 2 (95 %)
Shrinks with more harvested trees?	Yes, like $1/\sqrt{n}$	No
Shrinks when we add up many of *our* trees?	No (systematic)	Yes, partly (random)
How we carry it in Monte Carlo	one shared multiplicative draw, SD 1 %	one independent draw per tree, $\varepsilon \sim \text{Normal}(0, 0.357)$ on the log scale

3.5 Two more things the paper admits

- **Bias.** Averaged over the 58 sites the equation guesses about 5 % too heavy for ordinary trees and about 20 % too light for giants over 30 t. A lean is systematic: it stays however many trees we measure. For honesty we name it; we do not correct for it, because we do not know that it applies to young planted trees.
- **Tested range.** Every one of the 4,004 trees had $D \geq 5$ cm and grew in natural forest. Our 5-year-old, densely planted Miyawaki trees sit right at that edge, and the paper's smallest trees had its largest errors. Using an equation at the edge of its data is allowed; pretending the error is only σ is not. Say so in the report.

3.6 The sentence to remember

Chave et al. (2014) write that their method "provides estimates – not direct measurements – and model error should always be carefully examined and propagated through to the quantity of interest." Parts 5–7 do exactly that propagation.

Part 4 · One tree: are you certain the numbers you wrote down are right?

Your field sheet says tree #406: C = 25.1 cm, so D = 8.0 cm; H = 6.0 m. Before any equation, ask: if a classmate measured the same tree five minutes later, would they write the same numbers? If not, by how much would they differ? That difference is your **input uncertainty**, and it has to be put into words and then into a number.

4.1 Three ways to put a number on "not sure"

(a) Several people measured it. Tree #406 was measured by all 24 groups. The heights ran from 5.0 to 7.3 m (mean 6.14 m, SD 0.68 m, CV 11 %); the diameters from 7.5 to 8.5 cm (mean 7.98 cm, SD 0.26 cm, CV 3 %). The SD across people *is* the die for that measurement. Bootstrap it (Part 2.2). This is the best case and it is why the project has reference trees measured by every team.

(b) You measured it once, but you know your instrument and method. Then build the uncertainty from parts. Height by tape-and-angle: $H = d \cdot \tan\theta$ + eye height. With d = 10 m and θ = 30°, a 1° slip changes H by about 0.24 m (4 %); a 0.3 m slip in d changes H by 0.17 m (3 %); choosing a different twig as "the top" can change H by 0.5 m (8 %). Combine in quadrature: $\sqrt{(4^2 + 3^2 + 8^2)} \approx 9$–10 % for one careful reading. Diameter by tape: 1 mm reading on an 8 cm trunk is 1 %; the stick drifting 10 cm up or down a tapering stem adds 1–2 %; an oval stem adds a few percent. Together 2–3 %.

(c) You have nothing but experience: an educated guess. State it and defend it. Reasonable defaults for this project, which we use in Parts 6 and 7 whenever repeat readings are missing:

Input	Educated guess (95 % range)	As an SD for Monte Carlo	Why
Height H, tape-and-angle, measured once	±20 %	10 % of H	matches the 11 % spread seen across 24 groups; height is remote and computed
Diameter D, tape at 1.3 m	±5 %	2.5 % of D	matches the 3 % spread across groups; a contact measurement
Wood density ρ from a database	±20 %	10 % (log-normal)	four published values for Pongamia span 0.54–0.64; young wood is lighter than mature timber

A ±20 % range that covers 95 % of readings corresponds to an SD of about 10 % (Part 1.5: 95 % is ±1.96 SD). An educated guess is not cheating. It is an assumption, written down, that a reader can challenge and that you can replace with data later.

4.2 Height is much harder than diameter

Diameter is a **contact** measurement: you touch the stem, make one decision (where is 1.3 m?), and pull the tape tight. Different people get nearly the same number. Height is a **remote, computed** measurement: you never touch the top, you guess which twig is the top in a dense canopy, you read an angle, tape a distance and add your eye height. Each step has its own wobble, and tanθ stretches angle errors. Typical result: 2–3 % spread in D, 10–20 % in H. Yet D is squared in the equation, so a 3 % wobble in D becomes 6 % in AGB, while 10 % in H stays 10 %. Height still wins as the larger field error.

4.3 Random wobble or systematic lean? Ask for each input

- A different reader each time, a different stance, a different twig: **random**. Averages out over many trees.
- Everyone used the same stick with the mark at 1.25 m instead of 1.30 m; everyone added an eye height 10 cm too large; the tape counts bark as wood; the angle method under-reads leaning trees; the hidden true top in a dense canopy: **systematic**. Averages out over nothing.

For a single tree the distinction hardly matters: both widen the interval. For 25 trees (Part 7) it matters a great deal.

4.4 The full list of places where the number can go wrong

The number 23.5 kg CO_2 for tree #406 passes through a chain: tape and angle → D, H → ρ from a table → the equation (a, b) → the model's scatter → carbon fraction → CO_2 factor. Every link leaks.

#	Source	Where in the chain	Type	Rough size (1 SD, for one tree)	How to shrink it
1	Tape / angle-reader resolution	instrument	random	≈ 1 %	already fine
2	Person-to-person spread in D	input	random	2–3 % (→ 4–6 % in AGB)	same stick, training
3	Person-to-person spread in H	input	random	10–20 %	measure twice, cross-check groups
4	Oval stem, bark, lean, hidden top	tree shape	systematic for that tree	few %	note it on the sheet
5	Wood density ρ from a database	input	systematic (same for a species)	± 8–10 %, more for young wood	core a tree; use regional values
6	Equation parameters 0.0673, 0.976	model	systematic	< 1 %	nothing (needs more felled trees)
7	Model residual scatter σ = 0.357	model	random per tree	≈ 40 %	nothing for one tree; averages over many
8	Equation used outside tested range (D ≈ 5 cm, plantation)	model	systematic, unknown	unknown, possibly large	name it; compare with a second equation
9	Carbon fraction 0.47	conversion	systematic	± 5 %	species-specific value
10	C → CO_2 factor 44/12 = 3.67	conversion	exact	0 %	—

Do not memorise the numbers. Look at which rows are big: row 7 (the model's own scatter) dwarfs everything; row 3 (height) is the largest field item; rows 5, 6, 9 are small but never average away.

4.5 What "error in the inputs we want to predict from" means

The Chave equation needs three inputs: D, H, ρ. We measured two of them (with wobble) and looked the third up (with spread). The equation then *amplifies* or *damps* each wobble: D is squared (×2), H and ρ enter to the power 0.976 (×1). So, in percent, a wobble of 2.5 %

in D, 10 % in H and 10 % in ρ becomes about $\sqrt{((2\times2.5)^2 + 10^2 + 10^2)} \approx 15$ % in AGB before the model's own scatter is added. Part 5 lets the computer do this arithmetic by rolling dice.

Part 5 · Bootstrap and Monte Carlo applied to one tree

For tree #406 the honest answer is AGB ≈ 14 kg with a 95 % interval of 6.4 to 29 kg, a factor of two each way, and 92 % of that spread comes from the equation's own scatter. Interactive page: *Monte Carlo, One Tree*.

5.1 Step 1 — turn the 24 readings into dice (bootstrap)

From the 24 group readings: mean H = 6.14 m, bootstrap SE 0.13 m, 95 % interval 5.88–6.40 m; mean D = 7.98 cm, SE 0.05 cm. Two different things are now available, and you must choose the right one:

- If you want the AGB of tree #406 **as the class knows it** (24 readings pooled), the H die is narrow: SD 0.13 m (2 %). The class mean is well known.
- If you want the AGB **as one group would report it** (one reading), the H die is the spread of single readings: SD 0.68 m (11 %). This is the die for your own 25 trees, each measured once.

This guide uses the second, because it is your project's situation. We round it to the educated-guess default of 10 %.

5.2 Step 2 — roll all the dice (Monte Carlo)

The five dice and their shapes:

Die	Distribution used	Where it comes from
H	Normal, mean 6.0 m, SD 10 %	spread of single readings across groups
D	Normal, mean 8.0 cm, SD 2.5 %	spread across groups
ρ	Log-normal, median 0.60 g/cm³, SD 10 %	four database entries 0.54–0.64, plus young-wood doubt
a, b (0.0673, 0.976)	shared multiplicative draw, SD 1 %	refitting on resampled calibration data; Chave does not print SEs, so this is a generous guess
ε (model scatter)	Normal(0, 0.357) on the log scale, AGB × e^ε	residual standard error of Chave et al. (2014)

One iteration: draw H = 6.4, D = 7.9, ρ = 0.57, a = 0.0675, ε = −0.21 → AGB = 0.0675 × (0.57 × 7.9² × 6.4)^0.976 × e^(−0.21) ≈ 10.9 kg. Write it down. Repeat 20,000 times. The pile of 20,000 answers is the result.

5.3 Step 3 — read the pile, then roll one die at a time

Results of 20,000 draws for tree #406 (your numbers will differ slightly; that is the nature of the method):

Dice switched on	CV of AGB	95 % interval (kg)
H only	10 %	11.0 – 16.2
D only	5 %	12.3 – 14.9
ρ only	10 %	11.3 – 16.5
a, b only	1 %	13.3 – 13.9
model scatter ε only	37 %	6.7 – 27.6
all together	**40 %**	**6.4 – 29.0** (median 13.5)

Check by hand with the quadrature rule of Part 1.9: $\sqrt{(37^2 + 10^2 + 10^2 + 5^2 + 1^2)} = \sqrt{1595} \approx 40$ %. The computer and the formula agree. Notice what the arithmetic says: a *perfect* tape and a *perfect* laser (rows H and D set to zero) would reduce 40 % to $\sqrt{(37^2 + 10^2 + 1^2)} \approx 38$ %. Careful field work cannot fix the model's scatter; only averaging many trees can.

Converted: carbon = 0.47 × AGB → 6.4 kg (3.0–13.6); CO_2 = 3.67 × C → 23 kg (11–50). Note that we held 0.47 fixed; adding its ±5 % would widen the interval slightly.

5.4 Why the median, not the mean

The pile is lopsided (Part 1.7). Its median, 13.5 kg, sits at the deterministic answer 13.6 kg. Its mean is about 6 % higher, because e^ε with σ = 0.357 averages e^(σ²/2) = 1.066, not 1. Statisticians argue about which to report (the mean is the unbiased expected weight; the median is the typical tree). For this course: report the **median and the 95 % interval**, and say which you used.

5.5 How to write the result

Wrong: "AGB = 13.608 kg." Four digits claimed from a tape, a broken laser and a model with 40 % scatter.

Right: "AGB ≈ 14 kg (95 % interval 6.4–29 kg). Interval from 20,000 Monte Carlo draws carrying the group-to-group spread in H (10 %) and D (2.5 %), a 10 % log-normal spread in database wood density, 1 % on the fitted parameters, and the published model scatter (σ = 0.357). Dominated by the model's own scatter. Carbon fraction fixed at 0.47."

A number without an uncertainty is not a measurement. It is an opinion.

Part 6 · The Chave (2014) equation, term by term

The equation has five uncertain terms (a, b, ρ, D, H) plus the scatter ε. The original paper gives you a number for ε and the bias; it does not give you numbers for the others, so you must bring them from elsewhere or from an educated guess, and say which.

6.1 The general form

Written the way it was fitted, on the log scale:

```
\ln(AGB) = \ln a + b \cdot \ln(\rho \, D^2 \, H) + \varepsilon, \qquad
\varepsilon \sim \mathcal{N}(0, \sigma^2)
```

and back on the ordinary scale:

```
AGB = a \, (\rho \, D^2 \, H)^{b} \, e^{\varepsilon}
```

with a = 0.0673, b = 0.976, σ = 0.357 (Chave et al. 2014, Model 4, n = 4,004 trees, 58 sites). The term e^ε is the part most people forget. Dropping it gives a single number; keeping it gives the honest spread.

6.2 Each term and where its uncertainty comes from

Term	Meaning	What Chave et al. (2014) give	What is missing	Our educated guess / substitute (state it in the report)
a = 0.0673	scale: a tree is ≈ 7 % of its cylinder	the fitted value	its standard error	shared multiplicative SD 1 % (with 4,004 trees the SE is far below this)
b = 0.976	how weight grows with size (close to 1 = nearly proportional)	the fitted value	its standard error, and its correlation with a	included in the same shared 1 %
ρ (g/cm³)	wood density, from a database, not measured	nothing (they used site values)	spread between database entries; difference between mature timber and 5-year-old wood	log-normal SD 10 % per species; Chave et al. (2004) assume 10 %; the Global Wood Density Database gives a species SD you can use instead
D (cm)	diameter at 1.3 m, from tape circumference ÷ π	nothing (assumed exact)	your reading error	SD 2.5 % (±5 % range); or Chave et al. (2004): SD = 0.0062·D + 0.0904 cm for 95 % of trees and a gross error of 4.6 cm on 5 % of trees
H (m)	total height, computed from distance and angle	nothing (assumed exact)	your reading error	SD 10 % (±20 % range); Chave et al. (2004) assume 10 %; replace with the bootstrap SD of the reference trees when you have it

Term	Meaning	What Chave et al. (2014) give	What is missing	Our educated guess / substitute (state it in the report)
ε, σ = 0.357	one tree's distance from the line, log scale	**given**: σ = 0.357; CV ≈ 56 % averaged over sites; bias ≈ +5 % for ordinary trees, −20 % above 30 t	nothing	use as given; independent draw per tree
tested range	D ⩾ 5 cm, natural forest, no plantations	**given**	how wrong it is for 5 cm planted trees	cannot be quantified; name it, and compare with a second equation as a sensitivity test
carbon fraction 0.47	not in Chave; IPCC default	—	species value	±5 %, systematic
44/12 = 3.67	kg CO_2 per kg carbon	—	nothing, exact	0 %

6.3 The height-free version, for trees whose top you could not see

Chave et al. (2014) also fit a version without H, using an environmental index E (from the site's climate) in place of height:

```
\ln(AGB) = -1.803 - 0.976\,E + 0.976\,\ln\rho + 2.673\,\ln D - 0.0299\,(\ln D)^2 + \varepsilon
```

It has more fitted coefficients (each with its own, unprinted standard error) and a slightly larger scatter. The same logic applies: each coefficient is a die, E is a die (it depends on how well the climate maps describe Palakkad), and ε is the big die. If you use it, say so, and expect a wider interval, not a narrower one. Removing height from the equation does not remove the uncertainty that height carried; it moves it into the scatter.

6.4 A rule for every missing term

When the paper does not give an uncertainty for a term, do the following in order, and write down which step you reached:

1. **Measure it yourself** (repeat readings, bootstrap). Best.
2. **Find it in another paper** that measured the same thing (Chave et al. 2004 for D and H error; the wood density database for ρ).
3. **Make an educated guess**, choose it a little pessimistic (±20 % for a height measured once, ±5 % for a diameter), and test whether the final answer changes much if you double it. If it does not, the guess does not matter. If it does, say that it does.

What you may never do is set a term's uncertainty to zero because nobody gave you a number.

Part 7 · From one tree to a plot: 25 trees added together

Each of the 25 trees alone is uncertain by about ±40 % (a factor of two at 95 %), but their total is uncertain by only about ±12 %: 286 kg point estimate, 95 % interval 244–387 kg. Interactive page: *From 25 Trees to a Plot*.

7.1 The example plot

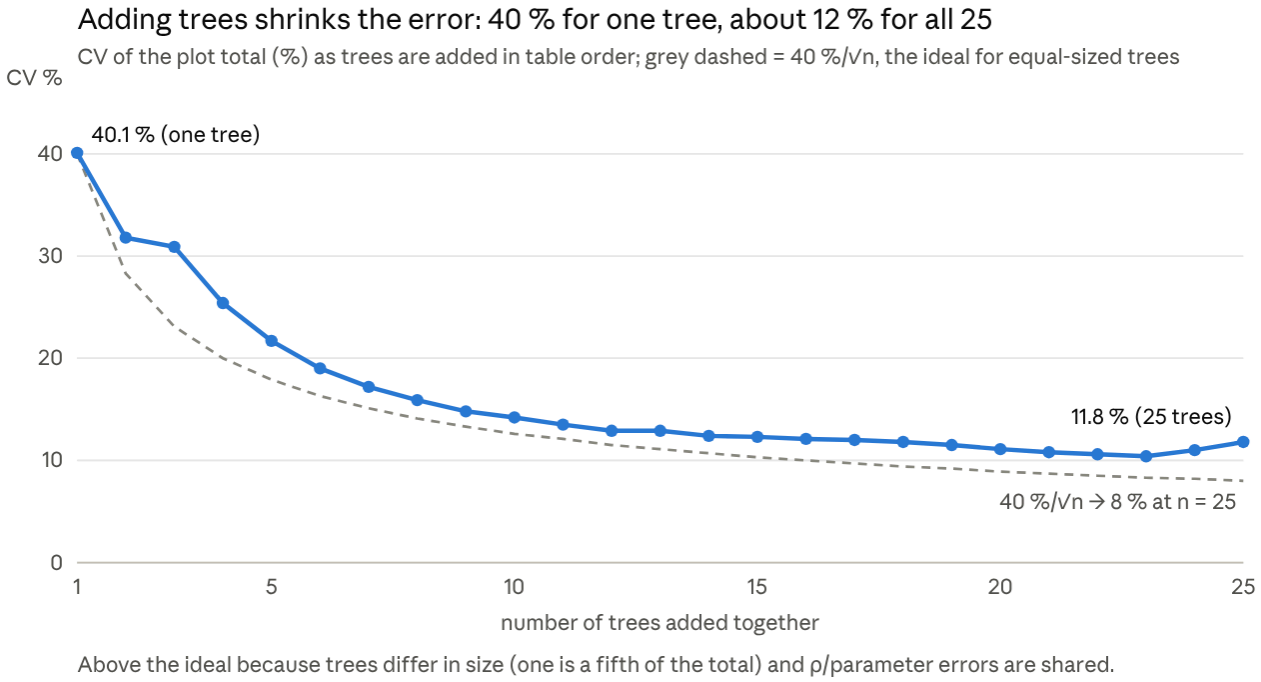
The species are the first 25 trees of group B1 from the class tree list. D and H are **illustrative** values typical of a 5-year-old Miyawaki plot (replace them with your own sheet). Wood densities are approximate species means from the Global Wood Density Database; look up your own species and record the source. Dice as in Part 6: H 10 %, D 2.5 %, ρ 10 % per species (shared by all trees of that species), parameters 1 % (shared by all trees), model scatter σ = 0.357 per tree.

Tree	Species	D (cm)	H (m)	ρ (g/cm³)	AGB point (kg)	95 % interval (kg)	CV
1	Terminalia bellirica	4.9	4.4	0.62	4.0	1.9 – 8.5	40 %
2	Phyllanthus emblica	3.0	3.3	0.70	1.3	0.6 – 2.7	40 %
3	Azadirachta indica	7.8	6.1	0.69	15.1	7.1 – 32.1	40 %
4	Terminalia bellirica	5.9	5.2	0.62	6.7	3.2 – 14.4	40 %
5	Prosopis cineraria	6.0	4.9	0.75	7.9	3.7 – 16.8	40 %
6	Simarouba glauca	7.9	6.8	0.39	9.9	4.6 – 20.8	40 %
7	Syzygium cumini	6.4	5.3	0.70	9.1	4.2 – 19.1	40 %
8	Psidium guajava	6.8	5.2	0.70	10.0	4.7 – 21.2	40 %
9	Terminalia arjuna	5.7	5.1	0.73	7.3	3.4 – 15.5	40 %

Tree	Species	D (cm)	H (m)	ρ (g/cm³)	AGB point (kg)	95 % interval (kg)	CV
10	Simarouba glauca	6.2	4.3	0.39	3.9	1.9 – 8.3	40 %
11	Terminalia arjuna	5.5	5.1	0.73	6.8	3.1 – 14.4	40 %
12	Syzygium cumini	6.0	5.0	0.70	7.6	3.5 – 16.0	40 %
13	Azadirachta indica	7.8	7.7	0.69	18.9	9.0 – 39.6	39 %
14	Prosopis cineraria	5.1	5.1	0.75	6.0	2.8 – 12.7	40 %
15	Bombax ceiba	4.0	3.6	0.33	1.2	0.6 – 2.5	40 %
16	Terminalia bellirica	4.2	4.3	0.62	2.9	1.3 – 6.1	40 %
17	Cleistanthus collinus	8.2	7.5	0.80	23.5	11.0 – 49.6	40 %
18	Swietenia mahagoni	4.7	4.4	0.60	3.6	1.7 – 7.6	40 %
19	Terminalia bellirica	5.5	4.4	0.62	5.0	2.3 – 10.6	40 %
20	Terminalia bellirica	7.5	6.7	0.62	13.8	6.4 – 29.2	40 %
21	Terminalia arjuna	6.9	5.3	0.73	10.9	5.1 – 23.0	40 %
22	Pongamia pinnata	4.8	4.9	0.60	4.1	1.9 – 8.7	40 %
23	Terminalia bellirica	7.7	6.8	0.62	14.7	6.9 – 31.4	40 %
24	Azadirachta indica	10.9	7.6	0.69	35.9	16.8 – 76.5	40 %
25	Cassia fistula	11.6	10.0	0.73	56.0	25.6 – 117.8	40 %
Total	13 species				**286**	**244 – 387** (median 303)	**12 %**

Every single tree has the same 40 % CV, because the model scatter dominates and it is the same for every tree. The total has 12 %. Nothing was measured better; the highs and lows of 25 independent scatters partly cancelled.

7.2 Why the total is better than any tree



Monte Carlo, 20,000 draws, 25 example trees of group B1 · default dice of Part 6 · computed for this guide

Three things to read off the curve:

1. **It falls like 1/√n at first.** Two trees: 32 %. Four trees: 25 %. Nine: 15 %. Averaging works exactly as the dice in Part 2.1 promised.
2. **It sits above the ideal 40 %/√n line.** The ideal assumes 25 equal trees. Ours range from 1.2 kg to 56 kg, and the largest tree alone is a fifth of the total. In effect the plot behaves like only about 12 equal trees, not 25. Adding the two biggest trees last (n = 24, 25) even nudges the CV back up. Rule: the error of a total is governed by the few trees that dominate it. Measure the big trees most carefully.
3. **It will not go to zero.** The shared dice (ρ per species, the equation's parameters, the carbon fraction, the +5 % lean) do not cancel between trees. Switch off the model scatter and the reading wobble entirely and about 4–5 % remains with 13 species; with a single species it would be about 10 %. More trees never remove a systematic error.

7.3 Median, mean and the point total

The point total (every tree computed with the plain equation) is 286 kg. The Monte Carlo median is 303 kg and the mean 306 kg. The 6 % gap is the log-normal effect of Part 5.4: a sum of 25 lopsided piles is itself nearly bell-shaped, so its median moves up to meet its mean, $e^{(\sigma^2/2)} = 1.066$ above the plain equation. All three sit comfortably inside the 244–387 kg interval. Report the median with the interval, and mention the point total so a reader can reproduce it on a calculator.

7.4 Dividing by an area: per hectare

The lecture quotes a typical Miyawaki planting density of about 3 trees per m², so 25 trees occupy roughly 8.3 m² (**assumption: replace with your plot's measured area**). Dividing by area does not change the percent uncertainty; it only changes the unit:

Quantity	Point	Median	95 % interval
AGB of 25 trees	286 kg	303 kg	244 – 387 kg
per m² (÷ 8.3 m²)	34 kg/m²	36 kg/m²	29 – 46 kg/m²
per hectare (× 10,000 / 8.3)	343 t/ha	364 t/ha	293 – 464 t/ha
carbon per hectare (× 0.47)	161 tC/ha	171 tC/ha	138 – 218 tC/ha
CO_2 per hectare (× 3.67)	592 t/ha	627 t/ha	506 – 800 t/ha

Two warnings belong next to that table in any report:

- **Scaling 8 m² to a hectare is an extrapolation.** The interval above carries only measurement and model uncertainty. It does not include the sampling uncertainty of *which* 8 m² you happened to measure. Chave et al. (2004) found that sampling error alone is about 10 % for a 0.25 ha plot and 5 % for 1 ha, and far larger for a few square metres. Pooling all 24 groups (600 trees, about 200 m²) is the honest way to approach a per-hectare figure.
- **Sanity-check the number.** 300–450 t/ha of above-ground biomass is what a mature tropical rainforest carries, not a 5-year-old planting. Either the density assumption is too high, the illustrative D and H are too generous, or the equation (tested on natural forest, D ⩾ 5 cm) overestimates small, crowded, planted trees. A result that fails a sanity check is a finding, not a failure: write it down and name the suspects. This is the kind of claim the AI-audit session taught you to test.

7.5 The error budget to put in your report

Source	One tree alone	Total of 25 trees	Shrinks with more trees?
Model scatter σ = 0.357	37 %	≈ 9 % (unequal sizes; 7 % if equal)	yes (random)
Reading wobble in H and D	≈ 11 %	≈ 3 %	yes (random)

Source	One tree alone	Total of 25 trees	Shrinks with more trees?
Wood density (13 species, 10 % each)	10 %	≈ 4 %	partly (cancels across species, not within)
Equation parameters	1 %	1 %	no
Carbon fraction (held fixed here)	(±5 %)	(±5 %)	no
Equation outside tested range, +5 % lean	not quantified	not quantified	no
Combined (quadrature)	**≈ 40 %**	**≈ 12 %**	

Name the biggest row. For one tree it is the model. For the plot it is still the model, but now the systematic rows are catching up, and only better data (a local wood density, a local equation) can lower them.

Part 8 · Assumptions, project checklist, exam practice, glossary, references

8.1 Every assumption this guide made, in one place

Assumption	Where	Why it is reasonable	What would break it
The 24 readings of a reference tree are a fair, independent sample of the measurement process	bootstrap, Part 2.2	different groups, different days, no copying	a shared wrong eye height; groups comparing numbers
Reading errors in H and D are bell-shaped with SD 10 % and 2.5 %	Parts 4–7	matches the spread across groups and Chave et al. (2004)	gross blunders (wrong tree, wrong unit) are not bell-shaped; clean them first
Wood density error is log-normal, SD 10 %, identical for all trees of a species	Parts 6–7	a database value is one number per species	within-species variation; young wood lighter than the table

Assumption	Where	Why it is reasonable	What would break it
Parameter error is 1 % and shared by all trees	Parts 3, 5–7	4,004 trees pin the line down	using a different equation fitted on 50 trees
Model scatter $\varepsilon \sim \text{Normal}(0, 0.357)$ on the log scale, independent between trees	Parts 3, 5–7	reported by Chave et al. (2014)	neighbouring trees in a crowded plot may share a shape; σ may be larger at $D \approx 5$ cm
Errors from different sources are independent of each other	quadrature, Monte Carlo	no mechanism links a tape slip to a database value	a group that reads both H and D high
The carbon fraction 0.47 and 44/12 are exact	Parts 5, 7	0.47 is the IPCC default; 44/12 is chemistry	species with 0.44 or 0.50 would shift the carbon figure by ±5 %
25 trees stand on 8.3 m² (3 per m²)	Part 7.4	typical Miyawaki density	measure the actual area of your block

8.2 Checklist for the project report

- [ ] Plot the H and D of the 10 reference trees, one bar per group. Is your group high, low or typical?
- [ ] Bootstrap the reference-tree readings: SD and 95 % interval of H and D. State the three bootstrap assumptions.
- [ ] For one of your 25 trees, run Monte Carlo one die at a time, then all together. Report the CV per source and the 95 % interval. Check with the quadrature rule.
- [ ] Sum your 25 trees with independent model-scatter draws and shared ρ and parameter draws. Report point total, median, 95 % interval, CV.
- [ ] Draw the error budget (one tree vs 25 trees) and name the biggest row.
- [ ] Convert to carbon and CO_2, and to per m² and per hectare with the measured area. Add the two warnings of Part 7.4.
- [ ] Write every number as estimate + interval + one line on where the interval came from + what was held fixed.
- [ ] Answer: which single change to the field protocol shrinks the interval most, and why is it not a better tape?
- [ ] Put the prompts you gave any AI tool in the appendix, and say which of its outputs you checked and how.

8.3 Exam-style questions (answers in brackets)

1. Twenty-four readings of a height have mean 6.0 m and SD 0.6 m. What is the standard error of the mean, and the 95 % interval? [SE = 0.6/√24 = 0.12 m; 5.76–6.24 m]
2. In the bootstrap, why are chips put back into the bag? [The 24 chips stand in for the whole process (the die); the same value must be allowed to appear twice, otherwise every resample is the same set.]
3. A tree's AGB has independent uncertainties of 37 % (model), 10 % (H), 6 % (D^2) and 10 % (ρ). What is the combined CV? What if the height were known exactly? [$\sqrt{(37^2+10^2+6^2+10^2)} \approx 40$ %; without H, ≈ 39 %. Field care barely helps for one tree.]
4. σ = 0.357 on the log scale. Between what factors does a single tree's true weight lie 95 % of the time? [e^(±1.96×0.357) = 0.50× to 2.0×]
5. Name the two kinds of model uncertainty and say which shrinks when more trees are *harvested* and which shrinks when more of *our* trees are *added*. [Parameter uncertainty: shrinks with more harvested trees, not with adding ours (systematic). Residual scatter: never shrinks with harvesting, partly cancels when ours are added (random).]
6. 25 equal trees each with 40 % random uncertainty: CV of the total? Why is the real figure for our plot higher? [40/√25 = 8 %; trees are unequal (effective n ≈ 12) and some errors are shared.]
7. The D of a tree is 3 % uncertain. How uncertain is D^2? [about 6 %: a power multiplies the relative error by the exponent.]
8. Give two systematic errors in this project and explain why measuring 1,000 trees would not remove them. [Database wood density; carbon fraction; the equation's +5 % lean; a stick marked at the wrong height. Each pushes every tree the same way, so the average keeps the push.]
9. Why does the Monte Carlo mean of a single tree sit above the plain-equation value? [Multiplying by e^ε with ε ~ N(0, σ²) has mean e^(σ²/2) = 1.066; the pile is lopsided; the median matches the plain value.]
10. A report says "CO_2 stored = 23.451 kg". Rewrite it correctly. ["CO_2 ≈ 23 kg (95 % interval 11–50 kg), from 20,000 Monte Carlo draws; dominated by model scatter; carbon fraction fixed at 0.47."]

8.4 Glossary

Term	Plain meaning
AGB	above-ground biomass: oven-dry weight of trunk, branches and leaves, in kg

Term	Plain meaning
Allometric equation	a formula predicting something hard to measure (weight) from easy measures (D, H, ρ)
Bias (systematic error)	an error that leans the same way every time; averaging does not remove it
Bootstrap	re-sampling your own data with replacement to see how much a statistic wobbles
CI, 95 %	the range that would contain the true value in 95 of 100 repeats of the method
CV	coefficient of variation: SD ÷ mean, as a percent
DBH, D	diameter at breast height (1.3 m), in cm; from tape circumference ÷ π
Log-normal	a lopsided pile that becomes bell-shaped when you take logarithms; multiplicative errors produce it
Mean, median	ordinary average; the middle value when sorted
Monte Carlo	drawing every uncertain input from its distribution, computing the result, repeating thousands of times
Quadrature	combining independent errors by square, add, square-root
Random error	an error with no pattern; averaging many readings shrinks it by √n
Residual scatter, σ	the vertical spread of real trees around the fitted line, in log units
SD	standard deviation: typical distance of one value from the mean
SE	standard error: SD of the mean = SD/√n
ρ (rho)	wood density, g/cm³, usually from a database

8.5 References

- Chave J. et al. (2014) Improved allometric models to estimate the aboveground biomass of tropical trees. *Global Change Biology* 20: 3177–3190. doi:10.1111/gcb.12629. Model 4: AGB = 0.0673 (ρD²H)^0.976, σ = 0.357, n = 4,004, 58 sites; height-free Model 7

with the index E.

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- Larjavaara M. & Muller-Landau H.C. (2013) Measuring tree height: a quantitative comparison of two common field methods in a moist tropical forest. *Methods in Ecology and Evolution* 4: 793–801.
- Lecture slides: *From simple measures to a tonne of CO_2* (Ecology & Environment 2026, lecture after the field visit); class tree list TreeSpecies_2026.xlsx.